Nicolas Nytko

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EDUCATION

University of Illinois at Urbana-Champaign Aug 2020 - Dec 2026 (Expected)
Doctor of Philosophy, Computer Science GPA 3.94 / 4.00
Advised by Luke Olson and Matthew West
Thesis: *Multilevel Solvers and Learning Methods for High-Performance Computing*

Master of Science, Computer Science
Thesis: *Learning Aggregates and Interpolation for Algebraic Multigrid*

University of Illinois at Urbana-Champaign Aug 2017 - Dec 2019
Bachelor of Science, Computer Science GPA 3.72 / 4.00
Grainger College of Engineering

Harper College Aug 2015 - May 2017
Transfer, Engineering Pathways GPA 3.59 / 4.00

PROFESSIONAL AND RESEARCH EXPERIENCE

Cerebras Systems Sept 2026 - Dec 2026
Intern, Advanced Technology Group *Sunnyvale, CA*

· Building high-performance numerical solevrs and scientific ML models on the Cerebras Wafer-Scale Engine

Center for Exascale-enabled Scramjet Design Aug 2024 - Sept 2026
Graduate Research Assistant *Urbana, IL*

· Developed JFNK implicit timestepping methods for nonlinear thermal radiation using discontinuous Galerkin spatial discretizations.
· Implemented lazily-evaluated multigrid solver for high performance computation on GPU clusters.
· Investigated use of element-level polytopic agglomeration for spatial coarsening.

Los Alamos National Laboratory Jun 2023 - August 2025
Graduate Research Assistant *Los Alamos, NM*

· Developed automatic code translation techniques for translating legacy structured PDE solvers to run on GPU accelerators, achieving $10 - 20\times$ speed-up on key computational bottlenecks.
· Developed structured auxiliary space methods for high-performance solvers on complex domains by learning diffeomorphic maps to structured domains using neural ODEs.
· Researched multi-resolution methods for learning problems on graph-structured data.

Enable Medicine May 2022 - Aug 2022
Computational Biology Intern *Menlo Park, CA*

· Researched numerical methods for computational biology spatial spillover correction.
· Explored both data-driven and algebraic methods for correcting cell biomarker error.

Groupon May 2021 - Aug 2021
Software Engineer Intern *Chicago, IL*

Epic Systems Corporation May 2018 - Aug 2019
Software Developer and Intern *Verona, WI*

PUBLICATIONS

Nytko N., MacLachlan S., Moulton J. D., Olson L. N., Reisner A., West M., *Unstructured to structured: geometric multigrid on complex domains via domain remapping*, Electronic Transactions on Numerical Analysis

Nytko N., Reisner, A., Moulton, J., Olson L., West M., *Teaching An Old Dog New Tricks: Porting Legacy Code to Heterogeneous Compute Architectures With Automated Code Translation*, International Journal of High Performance Computing Applications

Nytko N., Taghibakhshi A., Zaman T., MacLachlan S., Olson L. N., West M., *Optimized Sparse Matrix Operations for Reverse Mode Automatic Differentiation*, SIAM Journal on Scientific Computing

Zaman T., **Nytko N.**, Taghibakhshi A., MacLachlan S., Olson L., West M., *Generalizing Lloyd's algorithm for graph clustering*, SIAM Journal on Scientific Computing

Taghibakhshi A., **Nytko N.**, Zaman T., MacLachlan S., Olson L. N., West M., *MG-GNN: Multigrid Graph Neural Networks for Learning Multilevel Domain Decomposition Methods*, ICML 2023

Zaman T., **Nytko N.**, Taghibakhshi A., MacLachlan S., Olson L. N., West M., *Generalizing Reduction-Based Algebraic Multigrid*, Numerical Linear Algebra with Applications

Taghibakhshi A., **Nytko N.**, Zaman T., MacLachlan S., Olson L. N., West M., *Learning Interface Conditions in Domain Decomposition Solvers*, NeurIPS 2022

Silva M., Hieronymi P., West M., **Nytko N.**, Deshpande A., Chuang J., Hilgenfeldt S., *Innovating and modernizing a Linear Algebra class through teaching computational skills*, ASEE 2022

Silva M., Shaffer E., **Nytko N.**, Amos J., *A Case Study of Early Performance Prediction And Intervention in a Computer Science Course*, ASEE 2020

Nytko N., West M., Silva M., A Simple and Efficient Markup Tool to Generate Drawing-based Online Assessments, ASEE 2020

PRESENTATIONS & WORKSHOPS

41st Colorado Conference On Iterative and Multigrid Methods. *Hierarchical Graph Networks for Scalable Learning on Graphs.* June 25, 2026

22nd Copper Mountain Conference on Multigrid Methods. *Geometric Multigrid on Complex Domains via Mesh Remapping.* Apr 16, 2025

SIAM Workshop Series, UIUC. *Introduction to Machine Learning with PyTorch.* Nov 12, 2024

18th Copper Mountain Conference On Iterative Methods. *Learning sparse iterative methods through gradient-based optimization.* Apr 17, 2024

21st Copper Mountain Conference On Multigrid Methods. *Gradient-based optimization of sparse relaxation schemes.* Apr 17, 2023

17th Copper Mountain Conference On Iterative Methods. *Learning Aggregates and Interpolation for Algebraic Multigrid.* Apr 6, 2022

Hands-On With CSE, UIUC. *Building a Neural Network with PyTorch.* Apr 8, 2021

20th Copper Mountain Conference On Multigrid Methods. *A Supervised Learning Approach to Predicting Multigrid Convergence.* Mar 29, 2021

AE3 Lightning Symposium, UIUC. *Creating Drawing Questions in PrairieLearn.* Oct 3, 2019

HONORS AND AWARDS

Student Paper Winner: 22nd Copper Mountain Conference on Multigrid Methods — April 2025

Andrew & Shana Laursen Fellowship — Fall 2022 - Spring 2023

Outstanding Teaching Assistant — Spring 2021

Outstanding Course Assistant — Fall 2018

Dean's List, College of Engineering — Fall 2017, Spring 2018, Spring 2019

TEACHING EXPERIENCE

Graduate Teaching (UIUC)

CS 450 - Numerical Analysis (Fall 2023 - Spring 2024)

CS 357 - Numerical Methods I (Fall 2020 - Spring 2022)

MATH 257 - Linear Algebra (Computational Section Pilot) (Spring 2020)

CS 199 - Python for Data (Fall 2019 - Spring 2020)

Undergraduate Teaching (UIUC)

CS 357 - Numerical Methods I (Fall 2018 - Fall 2019)

CS 126 - Software Design Studio (Spring 2018)

ACTIVITIES

President, SIAM UIUC chapter — Fall 2024 - Spring 2026

President, Harper Society of Engineers — Spring 2017